Research Report

Revenue Forecasting

Best Practices, Methods and Telangana Applications

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This report reviews international and national revenue forecasting practices, analyses the major forecasting techniques (Time Series, Econometric, Macroeconomic, Consensus, and Microsimulation Models), and applies four methods to project Telangana's Tax Revenue and State's Own Tax Revenue for FY 2020-21 and FY 2021-22.

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Executive Summary

Revenue forecasting requires effective assignment and coordination of the Ministry of Finance (MoF), the research and planning unit of the Revenue Administration (RA), and other agencies involved in macroeconomic planning including the central bank, the national statistical agency, and the economic planning department. It is equally important to evaluate forecasting accuracy.

Fiscal marksmanship refers to the accuracy of a government's forecast of fiscal parameters such as revenues, expenditures, and deficits. If the gap between projected and actual tax revenues is large, it reflects poor fiscal marksmanship and undermines the credibility of the Budget.

International Forecasting Practices

Research shows that countries follow different statistical and econometric models for forecasting, based on assumptions about economic conditions, social and political forces, and data availability. Each method has distinct properties, accuracies, and costs. Techniques practiced internationally include:

- Time Series Methods (ARIMA, VAR, exponential smoothing)
- Economic Forecasts (base plus growth, baseline budgeting)
- Econometric / Macroeconomic Models (the most widely used approach in the UK, Italy, Germany, the Netherlands and Japan)
- Consensus Forecasting (used in the United States and Germany to depoliticise revenue estimates)
- Microsimulation Models (used in the Netherlands and US for income tax forecasting)

National Context: NIPFP Studies

This report explores three NIPFP working papers relevant to the Indian context:

- Kerala's Revenue Potential Projections for 1990–95: demonstrates the Tax System Approach for forecasting revenue potential.
- Tax Revenue Forecasts of Central, State, and Local Governments in India 2000–01 to 2004–05 (Eleventh Finance Commission): explains how different methods apply at different stages of revenue projection.
- Tax Revenue Efficiency of Indian States: The Case of Stamp Duty and Registration Fees — uses Stochastic Frontier Analysis (SFA) to measure tax capacity and tax effort across 16 major states.

Telangana Application

Four forecasting methods were applied to project Telangana's Tax Revenue and State's Own Tax Revenue (SOTR) for FY 2020-21 and FY 2021-22 using data from FY 2014-15 to FY 2019-20. The methods applied were: Tax Buoyancy/Elasticity, Macroeconomic (Logarithmic Regression), ARIMA, and Consensus Forecasting (Multiple Linear Regression).

Consensus Forecasting produced the most accurate projections, with the smallest forecast errors in both Tax Revenue and SOTR. Tax Buoyancy and Logarithmic Regression systematically overestimated revenues; ARIMA systematically underestimated.

1 Introduction

Revenue forecasting is key to successful budgeting in the public sector. Forecasting is the process of making predictions of the future based on past and present data, most commonly by analysis of trends. A commonplace example is the estimation of some variable of interest at some specified future date.

1.1 Context

Solid analytics must be in place for governments to make informed decisions on budgeting, taxes, and expenditures. A tax reform strategy should prioritise good revenue forecasting, analysis of the potential of the tax system, impact of changes in tax policy, and analysis of tax expenditures. The importance of tax analysis and revenue forecasting for proper budget management, particularly for the medium-term fiscal framework, cannot be overemphasised.

Four types of quantitative models are available for forecasting:

- Unconditional time-series analysis that includes trends and growth factor analysis. This model is unconditional in the sense that forecasts are based only on past revenue data and use econometric techniques.
- Macroeconomic or GDP-based modeling that uses a time series of tax revenues and their bases. This model relies on the link between the growth of revenue and the growth of tax bases or GDP. Data requirements are limited to time series of tax revenues and the corresponding tax base (consumption, income, imports, or simply GDP).
- Microsimulation models are based on tax returns of individual taxpayers or transactions, which are then aggregated. These models also enable estimation of the distributional effect of a given policy change. They are data-intensive and use microsimulation modeling techniques.
- Revenue receipts modeling that monitors and projects short-term receipts by type of tax or aggregate revenue. This is particularly useful for setting monthly or quarterly collection targets. Data requirements are simple and limited to monthly receipts in the current and prior years.

Once revenue is forecast to a reasonable degree of certainty, the forecast is compared with actual tax collections, yielding a measure of the tax gap. This analysis can be conducted by type of tax.

1.2 Features of a Revenue Forecasting System

A sound revenue-forecasting system has the following features:

- Dedicated, full-time qualified staff. Building revenue-forecasting expertise requires time and skill, particularly staff with proficiency in tax economics and econometrics. Capacity-building programs should be systematic and frequent.
- Ability to anticipate data needs and construct suitable databases. Policy makers may propose changes in tax policy whose revenue impacts cannot be estimated accurately on existing data. Tax returns and survey forms may need to be modified or redesigned.
- Adequate hardware and software. Staff should be equipped with both state-of-the-art hardware and cutting-edge software for statistical modeling and data management.

This report first outlines the purposes of revenue forecasting; second, discusses the basic concepts and issues involved with revenue measurement, estimation and growth; third, describes the types of models used for tax revenue forecasting and error analysis; fourth, explains the different models practiced across the world; and finally, discusses the organisational arrangements needed to support revenue forecasting.

2 Purposes and Importance of Revenue Forecasting

- Assessing the revenue impacts of policy changes. This includes changes in tax policies, non-tax policies (trade, industrial, financial regulations), and structural changes.
- Appraising the revenue impacts of economic changes or shocks. This includes GDP growth, devaluation, inflation, trade patterns, political developments in neighbouring countries, and weather events such as prolonged drought.
- Measuring tax capacity and tax effort. This includes measuring the quantity and quality of revenue forecasts, the potential tax revenue of the economy, and the degree to which that potential has been achieved.
- Monitoring and measuring the performance of the revenue department. This provides a benchmark for monitoring collection performance and tax effort, which can be linked to bonus schemes.
- Strengthening tax administration. This includes identifying improvements and expanding the tax base under existing tax legislation.
- Determining the institutional setup. Whether the Tax Policy Unit (TPU) or the tax administration should prepare the forecast must be determined.
- Maintaining the database. The quality of revenue forecasts is only as good as the quality of data. Data come from the tax administration, statistical bureau, central bank, Ministry of Finance, and Ministry of Economy, as well as trade or business organisations. The TPU should maintain and update the database continuously.
- Coordinating closely with economic ministries, government agencies, and business organisations. Coordination is essential to producing accurate economic forecasts. GDP is the source of all tax revenues, and an accurate and timely GDP forecast is crucial.

3 Basic Concepts of Revenue Measurement, Estimation and Growth

3.1 Measures of Revenues

A core issue in revenue forecasting is what accounting measures of revenue are being estimated. All governments maintain cash accounts and need revenue collection estimates on a cash basis, but an increasing number are maintaining accounts on a modified or full accrual basis.

In forecasting revenues, changes in economic conditions are expected to operate directly through changes in tax liabilities and only later through changes in tax collections based on those liabilities. For example, if a tax base grows due to real economic growth, then the legislated revenue liability on the expanded base arises first, and the revenue authorities subsequently attempt to collect as much of the increased liability with minimum delays. Year-to-year fluctuations in collections may also reflect changes in compliance and collections of outstanding tax liabilities from earlier years.

3.2 Determinants of Taxes

Tax revenue forecasting is based on forecasting the determinants of assessed revenues. The starting point of any tax revenue forecast is an understanding of the determinants of tax assessments in a tax period. The growth in tax revenues from one period to the next depends on changes in these determinants.

Different types of tax forecasting models assume that many determinants remain constant between periods and focus only on specific or major determinants, such as the real growth in the size of the economy. The choice of simplifying assumptions depends on the ability of a model to accommodate or estimate the effects of different determinants and/or the availability of data on the tax and its determinants.

When tax structures remain relatively stable over time, revenue forecasting can largely bypass the need to focus on behavioral responses to price effects arising from tax rate changes. Behavioral modeling is necessary to estimate:

- Changes in the tax revenues or the burden of the tax.
- The incidence of the tax burden.
- Excess burden or economic efficiency costs of taxation.
- Transaction costs from compliance with and administration of the tax change.

3.3 Revenue Growth

Governments have a basic interest in their revenue performance because they need to know whether they can finance a stable stream of public services over time. Two basic measures of revenue performance are the following:

Revenue Buoyancy: A simple measure of the growth rate of actual revenue collections over some span of years relative to the growth in the economy (usually GDP) over the same period. Buoyancy shows whether the overall revenue collection as a share of the economy is rising, falling, or remaining steady, depending on whether the buoyancy is greater than, less than, or equal to 1.

$$E^{bTy} = (\Delta T^b / T^b) \div (\Delta Y / Y)$$

Where: E^{bTy} = Buoyancy of tax revenue to income; T^b = Total tax revenue; ΔT^b = Change in total tax revenue; Y = Income/GDP; ΔY = Change in income

Revenue Elasticity: A more restrictive but more useful measure for revenue forecasting. It assumes that revenue growth occurred with no changes in tax policy or the nominal tax structure over the observed period. If elasticity is less than unity, it implies revenues are expected to decline as a share of the economy and that tax policy changes are required.

$$E^{Ty} = (\Delta T / T) \div (\Delta Y / Y)$$

Where: E^{Ty} = Elasticity of tax revenue to income; ΔT = Change in tax revenue; ΔY = Change in income/GDP

4 Time Horizon of the Forecast

Fiscal projections are generally divided into short-term, medium-term, and long-term forecasts. The generation of budgetary forecasts at different horizons is mainly driven by the specific objectives of the analysis.

4.1 Short-Term Revenue Forecasting

Short-term budgetary forecasts normally cover a time period of up to one year. They are mostly prepared by government institutions for budgeting and management purposes. Monthly and quarterly cash figures on individual revenue and expenditure items, as well as on net borrowing, provide useful indicators of the most recent budgetary developments. They serve as early warning indicators and provide timely information in case actual developments deviate significantly from projected annual paths.

4.2 Medium-Term Revenue Forecasting

Medium-term budgetary forecasts cover a time period of 2 to 5 years. There is a compelling need to ensure that medium-term forecasts are in line with the projected evolution of relevant macroeconomic bases. Failure to ensure consistency risks introducing significant forecast errors in cyclically sensitive revenue items and ultimately in government net borrowing and gross debt.

Fiscal forecasts contribute to highlighting the fiscal adjustment needed to fulfill government targets and could be based on a prudent policy scenario, which takes into account only well-specified and credibly announced policy measures.

4.3 Long-Term Revenue Forecasting

Long-term budgetary forecasts covering a period beyond 5 years and up to several decades aim to estimate budgetary pressures stemming from population aging, particularly in pension-related expenditure and healthcare. They allow for an assessment of fiscal sustainability and help identify potential needs for structural reforms.

Three main methodological approaches to fiscal sustainability assessments based on long-term forecasts are:

- Sustainability gap indicators — evaluate by how much taxes should be increased, or expenditure reduced, to keep the debt ratio constant by the end of the forecast horizon or to fulfil the government's intertemporal budget constraint.
- Generational accounting — adopts a microeconomic perspective to estimate the financial burdens imposed on different age groups and generations.
- General equilibrium models — more difficult to build and understand; based on modelling assumptions that can be contested but offer more nuanced and internally consistent policy conclusions.

5 Data Requirements for Revenue Forecasting

The quality of a revenue forecast is fundamentally determined by the quality and completeness of its underlying data. The following data categories are essential for building a credible revenue forecasting system:

5.1 Tax Administration Data

- Detailed taxpayer-level assessment and collection data from the Revenue Administration (RA), including returns filed, taxes assessed, taxes collected, and refunds issued.
- Disaggregation by tax type (income tax, GST, excise, stamp duty, etc.) to enable tax-specific modeling.
- Time series of monthly and quarterly receipts to support short-term monitoring models.

5.2 Macroeconomic Data

- GDP (nominal and real) and GSDP at the state level, disaggregated by primary, secondary, and tertiary sectors.
- Consumption, investment, import, and export data from the National Accounts.
- Inflation indices (WPI, CPI), interest rates, and exchange rates as independent variables in macroeconomic models.
- Employment and wage data from PLFS and the Economic Census.

5.3 Administrative and Policy Data

- Current tax rate structures, exemption thresholds, and applicable deductions for each tax head.
- Records of discretionary tax policy changes (rate changes, base expansions, new levies) by effective date.
- E-filing data, audit outcomes, and compliance statistics from the RA.

5.4 External and Survey Data

- Business tendency and consumer confidence surveys from chambers of commerce and RBI.
- Commodity price data for states with significant mineral or agricultural revenue exposure.
- Demographic data (population projections, age-cohort distribution) for long-term forecasting models.

6 The Forecasting Steps

A structured, seven-step process for revenue forecasting ensures methodological rigour and comparability of estimates across time periods.

Step 1: Selection of Forecast Period

This step involves selection of the period over which budget data are examined. It depends on: (a) the availability and quality of data; (b) the type and sustainability of the revenues or expenditures to be forecasted; and (c) the degree of accuracy sought. Central government gross revenue forecasts typically look one year ahead, while ministries such as Education and Health may look 10–20 years ahead for infrastructure and demand planning purposes.

Step 2: Examination of Data for Stationarity

This step involves examination of the data for structural characteristics such as trends and rates of change. The main concern is to identify evidence of:

- Stability (stationary behaviour)
- Non-stationary non-linear paths — exponential growth, quadratic paths, structural breaks, and seasonal or cyclical variation

A practical test: divide the data into 5 sequential parts. If the mean and variance of each part are approximately equal to those of the other parts, the series is stationary. If not, it is non-stationary and requires special treatment before forecasting.

Step 3: Adoption of Assumptions

Government fiscal policy is affected by economic, social, and political forces. Forecasting therefore requires explicit assumptions about how revenues are affected by: (a) changing economic conditions; (b) changing population size and citizen demand; (c) changing government policies; and (d) changing administrative procedures.

Step 4: Selection of Forecasting Methods

The method selected depends on the nature and type of revenue:

- Qualitative and judgmental methods are needed for highly uncertain revenues, including new revenue sources, grants, and asset sales.
- Quantitative methods are appropriate for revenues based on greater certainty, such as income taxes and sin taxes (alcohol, tobacco).
- When in doubt, the simplest appropriate method should be preferred, subject to evaluation in Steps 5 and 6.

More than one method may be applied and the results averaged to reduce individual model risk.

Step 5: Evaluation of Estimates

Evaluation of estimates requires assessment across two dimensions:

- Validity: validation of assumptions about the revenue source — economic, population (demand), administrative, and political assumptions.

- Reliability: sensitivity analysis by varying key parameters. If small parameter changes produce large changes in estimates, the projection has low reliability and should be treated with caution.

Step 6: Monitoring of Outcomes and Comparison with Forecasts

Actual revenue collections are observed and compared against estimates. The extent of deviation or error is measured and used to assess forecast accuracy. A forecast should be unbiased in the sense that the expected value of the deviation of actuals from the forecast should be zero. Forecast accuracy is related to data stationarity: errors tend to be larger near economic turning points (non-stationary behaviour) and smaller when the economy is growing near its trend (stationary behaviour).

Step 7: Updating the Forecasts

In the final step, forecasts are updated whenever assumptions must be changed. Updating is necessary when conditions in the economy are changing with respect to: (a) economic forces; (b) population and demand; (c) administrative arrangements; and (d) political developments.

7 Forecasting Techniques

7.1 Time Series Methods

Time series forecasting of revenues is based on the assumption that patterns in the historical data of a series can be used to project future revenues. The method relies on the concepts of trend, cycle, season, and random change.

7.1.1 Distinguishing Trend, Cycle, Season, and Noise

A time series is typically decomposed into four components:

Trend

Trend is the general direction of change of data over time. A series increasing over time has a positive (non-stationary) trend; one decreasing has a negative trend; one with no directional change is stationary. Identification of trend requires a series long enough for patterns to be discernible. The general rule: fewer than 50 observations is a ‘short’ series requiring special methods; most time series methods can be applied with 50 or more observations.

Cyclical

Cyclical fluctuations in a revenue time series result from business cycles. They represent the medium-term results of fluctuations in capacity utilisation and are the focus of medium-term expenditure frameworks. Introduction of VAT and user fees in health and education have made revenue flows significantly dependent on economic cycles.

Seasonal

Seasonality refers to revenue cycles that change within a year. If a time series tends to vary over the course of a year in response to systematic phenomena (dry/wet season transitions, school holiday periods, annual filing deadlines), it has a seasonal component. Seasonal fluctuations can be additive (size not related to the trend level) or multiplicative (size related to the trend level).

Random Noise

Randomness refers to unexpected events that distort otherwise stable trends — often called ‘noise’ or ‘shocks’ (e.g., natural disasters affecting tax flows). Removal of noise is called ‘smoothing’ or ‘filtering’. When noise is removed from a series, the other components become more visible and amenable to forecasting.

7.1.2 Assumptions

- Random error terms are normally distributed.
- There are dependable correlations between the variable to be forecast and other independent variables.
- Past patterns in the variable to be forecast will continue unchanged into the future.
- The data do not exhibit a trend (or the trend has been removed prior to modelling).

7.1.3 Limitations

- Conclusions drawn from time series analysis are not always perfect or reliable over long horizons.
- Various factors affecting fluctuations in a series cannot be fully adjusted within a time series framework.

- Factors influencing the time series may not remain stable over extended periods, making long-range forecasts unreliable.
- An increasing trend may be due to population growth rather than genuine revenue performance, requiring appropriate normalisation.

7.2 Economic Forecasts

Economic forecasting is the process of attempting to predict the future condition of the economy using a combination of important indicators. It involves building statistical models with inputs of several key variables — typically in an attempt to forecast future GDP growth. Primary economic indicators include inflation, interest rates, industrial production, consumer confidence, worker productivity, retail sales, and unemployment rates.

7.2.1 Assumptions

- People have rational preferences among outcomes that can be identified and associated with a value.
- Individuals maximise utility (as consumers) and firms maximise profit (as producers).
- People act independently on the basis of full and relevant information.

7.2.2 Limitations

All data provided must be complete and accurate for the analysis to be successful. Economic models primarily use mathematical or quantitative analysis where accuracy is critical. Qualitative analysis, sometimes incorporated in economic models, is known for lacking precision. Once a model is constructed, information is checked and updated as needed — but the model’s outputs are only as good as its inputs.

7.3 Econometric Models

An econometric model is one of the tools economists use to forecast future developments in the economy. Econometricians measure past relationships among variables such as consumer spending, household income, tax rates, interest rates, and employment, then try to forecast how changes in some variables will affect others. The Ordinary Least Squares (OLS) technique is the most popular method of performing regression analysis, because in standard conditions it produces optimal results.

7.3.1 Assumptions

- Model parameters are linear (regression coefficients do not enter as exponents).
- Values for independent variables are derived from a random sample and contain variability.
- Explanatory variables do not have perfect collinearity.
- The error term has zero conditional mean (average error is zero at any specific value of the independent variable).
- The model has no heteroskedasticity (variance of the error is constant regardless of the independent variable’s value).
- The model has no autocorrelation (the error term does not exhibit a systematic relationship over time).

7.3.2 Limitations

Econometrics is criticised for the perceived widespread methodological shortcomings of its analyses. The proliferation of econometric techniques has been disproportional to our understanding of the economy, which can undermine the policy purpose for which econometrics was developed. Models must be evaluated as tools to advance knowledge, with the ultimate objective of making sound policy decisions.

7.4 Macroeconomic Model

A macroeconomic model is an analytical tool designed to describe the operation of a country's or region's economy. These models are usually designed to examine the comparative statics and dynamics of aggregate quantities such as total output, total income, employment levels, and price levels. They are the basic workhorse models used by most governments to explain and forecast revenues over the medium term (one to five years) in terms of changes in macroeconomic aggregates such as real GDP, consumption, imports, and major price indexes.

7.5 Consensus Forecasts

Consensus forecasting is a technique that develops a forecast for a variable of interest by incorporating input from multiple sources and parties. When undertaking a consensus forecast, input is sought from experts who have a stake in the variable being forecasted. The process is more time and resource consuming than traditional forecasting approaches; however, using information from multiple people and sources allows a more robust perspective to be developed and reduces the influence of any single model's biases.

In the public sector, consensus forecasting is used in part to remove politics from the forecasting process and to prevent politicians from inflating revenue estimates, particularly in election years.

7.5.1 Theory of Consensus Forecasting

Consensus forecasts are predictions created by combining several separate forecasts, often developed using different methodologies. Also known as combining forecasts, forecast averaging, or model averaging (in econometrics), and as committee machines or ensemble averaging (in machine learning). Applications range from weather forecasting to predicting annual GDP growth or sector-level tax revenues.

7.5.2 Pooling Procedure and Forecast Analysis

To test for bias of consensus forecasts and to analyse forecast revisions, forecasts for each variable are pooled across horizons. This approach enables more powerful econometric tests than the traditional procedure which looks at forecasts separately for each horizon. The pooling procedure is particularly suited for datasets comprising forecasts over 24 horizons for each year but covering only a small number of target years.

Incorporating target year-specific shock variances improves econometric tests to analyse forecast revisions. This source of variability is important for monthly errors of GDP growth and inflation forecasts. The pooling approach can be used to conduct sequential tests to detect forecast biases for increasing horizons, providing a comprehensive picture of forecast performance.

7.6 Microsimulation Models

Microsimulation models are based on calculating taxes from the tax returns of individual taxpayers or transactions and then aggregating the results. A model contains a tax calculator that applies all tax rules to the tax information of each return and then aggregates tax liabilities across all relevant returns, cross-tabulating results as needed. Such models can handle complex tax logic, tax schedules, tax losses, credits, and inter-period carryovers.

These models are particularly powerful for the analysis of revenue, incidence, and efficiency effects of tax policy changes. They can also be used to forecast taxes if the tax information of each return in a model sample database can be projected into future tax periods on the basis of assumptions about economic growth, inflation rates, exchange rates, and compliance or enforcement changes. Their use depends critically on having computerised data files of information from a representative sample of tax returns.

8 Importance and Applications of the Models

The table below compares the five primary forecasting methodologies across importance, key models, and applications:

Table 1: Comparison of Forecasting Methodologies

Methodology	Importance	Key Forecasting Models	Applications
Time Series	Understand underlying structure enables prediction, monitoring, and control of fiscal outcomes.	ARIMA VAR Multivariate time-series Empirical Bayesian Deseasonalized SES Damped trend	Economic forecasting Sales forecasting Budgetary analysis Stock market analysis Yield projections Inventory studies Census analysis
Economic Forecasts	Helps policymakers make better decisions can forecast recessions enabling expansionary fiscal responses.	Economic base analysis Base plus growth Baseline budgeting	GDP growth Inflation House prices Exchange rates Population growth
Econometric Models	Comprehensive macro-relationship models used by central banks to evaluate and guide policy.	Macroeconomic model Simple/Multiple Linear Regression Log-regression	Macro indicator forecasting Immigration impact Minimum-wage effects Revenue response to marketing Cap-and-trade policy impact

Source: Compiled from Jenkins, Kuo & Shukla (2002); Congressional Budget Office (2007); and primary literature.

9 Confidence Intervals around Economic and Revenue Forecasts

9.1 Measures of Uncertainty around Economic Forecasts

Forecasts are subject to inherent uncertainties. Generally, these uncertainties tend to increase as the forecast horizon lengthens. Forecast errors (the differences between forecasts and outcomes) can arise for a range of reasons — for example, differences between the assumed path of key variables and outcomes, changes in the relationships between different parts of the economy, and unexpected domestic or global events.

Confidence intervals seek to illustrate that there is a range of plausible outcomes around any forecast. They are based on observed historical patterns of forecast errors and serve as a guide to the degree of uncertainty. In order to measure forecast error and ensure maximum accuracy, three techniques are commonly used:

- Root Mean Squared Error (RMSE) — the most reliable measure
- Mean Absolute Percentage Error (MAPE)
- Mean Absolute Error (MAE)

9.2 Revenue Forecast Errors

Evaluating forecast accuracy is a crucial element of forecasting procedures. It is relevant for monitoring purposes and allows for improved forecasts by learning from past errors. Sizeable, systematic, or biased forecast errors in certain items would allow fiscal analysts to identify weaknesses in their forecasting procedures in terms of methods, discussions, or decision-making processes.

9.2.1 Bias

If the costs of forecast errors were symmetric, forecasts should present no systematic bias (i.e., on average the forecast error should not differ significantly from zero). However, a government may tend to overestimate a deficit when the loss of an underestimation is greater. A second source of bias arises from systematically omitting relevant economic variables or from errors in fiscal variables induced by systematic errors in economic variables (output gap, price variables, GDP volatility) through estimated tax/spending elasticities.

Unbiasedness can be formally tested by regressing the budgetary outcome on the forecast:

$$R_t = \alpha + \beta F_t + u_t$$

Where: R_t = budgetary outcome; α = constant; F_t = forecast; u_t = error term. The null hypothesis of unbiasedness requires $\alpha = 0$ and $\beta = 1$.

9.2.2 Efficiency

The key for an assessment on efficiency or rationality lies, firstly, in the available information at the time the forecast was elaborated (data, policy measures), and secondly, in the use of this information (methods, procedures). Efficiency is investigated through:

(i) the presence of serial correlation in a time series of errors

(ii) the correlation of the forecast errors with information available at the time the forecast was performed.

9.2.3 Accuracy

"Accuracy," refer to two aspects of the forecast as compared to the actual outcome: (i) how close both are in quantitative terms and (ii) the capacity of the forecast to predict the direction of change in the final outcome. The accuracy of a set of forecasts can be considered using standard measures such as the Mean Error, Mean Absolute Error, or Root Mean Squared Error (RMSE), or with directional sign tests.

9.3 Causes of Forecast Errors

- Policy errors: due to errors in the course of fiscal policy, owing to the implementation of new or not-yet-announced fiscal policy measures or cancellation of previously announced measures.
- Economic errors: those explained by wrong forecasts of macroeconomic variables used in budget projections (GDP, interest rates, inflation).
- Technical errors: due to other factors, including behavioral responses or model mis-specification on the fiscal side.

9.4 Fiscal Marksmanship

Fiscal marksmanship essentially refers to the accuracy of the government's forecast of fiscal parameters such as revenues, expenditures, and deficits. If the difference between what the government projected as likely tax revenues in the Budget and the actual figures a year later is large, it reflects poor fiscal marksmanship.

In the Indian context, this term gained prominence after Raghuram Rajan, then Chief Economic Advisor, stressed on fiscal marksmanship in the Economic Survey for 2012-13, defining it as "the difference between actual outcomes and budgetary estimates as a proportion of GDP."

9.4.1 Why Does Fiscal Marksmanship Matter?

- The salience of Budget numbers lies in their credibility.
- The central purpose of publicly disclosing the Budget in a democracy—and seeking legislative approval, is to make policymaking and governance transparent and participatory.
- Everyone knows Budget numbers are forecasts and as such unlikely to tally exactly with actuals, but there is an underlying belief that they are based on genuine calculations.
- If fiscal forecasts turn out to be way off the mark repeatedly, it will undermine the credibility of the Budget presentation itself.

9.4.2 The Theil's Index

The Theil's Index (Theil 1958) is a methodology used to assess forecast accuracy. The range of U_1 is from zero to 1. A perfect forecast corresponds to $U_1 = 0$ (where $P_{\square} = A_{\square}$). U_1 equals 1 when either all predicted values are zero or all actual values are zero. However, U_1 has a known defect: forecasts with similar errors at different distances from the origin yield very different values of U_1.

A revised version (Theil 1966), U_2, is not bounded above but retains the lower bound of 0. A perfect forecast also corresponds to $U_2 = 0$. A more rigorous index, U_3, uses lagged values ($Q_{\square} = P_{\square} - P_{\square-1}$ and $a_{\square} = A_{\square} - A_{\square-1}$) and is also not bounded above.

10 International Revenue Forecasting Practices

The following section examines the revenue forecasting approaches adopted by six major economies, highlighting the institutional arrangements, model types, and key methodological features of each. A comparative summary is provided in Table 2.

Table 2: International Revenue Forecasting Practices — Comparative Summary

Country	Primary Approach	Institutional Arrangement	Key Methodology
United States	Baseline Budgeting	CBO, OMB, FRB, CEA, SSA, BEA	Quantitative (time-series, regression) + judgement; consensus in public sector to depoliticise forecasts
United Kingdom	Macroeconomic Model (OBR)	Office for Budget Responsibility (OBR) + HM Revenue & Customs	Three-group model: accounting identities, behavioural equations, technical relationships
Italy	Macroeconomic Models	Parliamentary Budget Office (UPB); Italian Treasury	ITEM (371 variables), IGEM (DGE), QUEST III (DSGE), MACGEM-IT (CGE)
Australia	Base Plus Growth	Treasury	Last known outcome as base; growth rates applied; supplemented by survey data and business liaison
Germany	Macroeconomic Models	Ministry of Finance; Working Party on Tax Revenue Forecasts (independent)	Three-arc annual cycle; iterative-analytic approach; microsimulation for discretionary measures
Netherlands / Japan / New Zealand	Econometric Models	CPB (NL), Ministry of Finance (JP), Treasury (NZ)	Regression analysis of revenue-indicator relationships; micro-simulation for income taxes; trend-extrapolation for minor taxes

Sources: Congressional Budget Office (2007); OBR (UK); Italian Draft Budgetary Plan 2020; Australian MYEFO 2019-20; German Ministry of Finance; CPB Netherlands.

10.1 The United States | Baseline Budgeting

Federal budget forecasts are generally associated with macroeconomic data, particularly unemployment, inflation, and GDP growth. At least six federal government entities make revenue, expenditure, deficit, and debt forecasts:

- The Council of Economic Advisors (CEA)

- The Office of Management and Budget (OMB)
- The Federal Reserve Board (FRB)
- The Congressional Budget Office (CBO)
- The Social Security Administration (SSA)
- The Bureau of Economic Analysis (BEA)

Most studies find that budget-year forecasts are relatively accurate, but there is evidence of optimism, particularly in OMB forecasts. Optimism can be defined as a forecast that leads to an underestimated deficit — often arising from underestimation of unemployment or inflation, or overestimation of GDP growth.

10.1.1 CBO's Baseline Projections (as of January 2020)

CBO's spring 2020 budget projections are based on the economic forecast the agency developed in January 2020, incorporating legislation enacted through 6 March 2020 and various technical adjustments based on new information. These projections did not account for changes to the economic outlook arising from the novel coronavirus public health emergency.

Table: Key Projections in CBO's March 2020 Baseline — as a Percentage of Gross Domestic Product

Revenue Category	2020	2021	Projected AnnualAverage 2022–2025	Projected AnnualAverage 2026–2030
Individual income taxes	8.1%	8.3%	8.4%	9.4%
Payroll taxes	5.9%	5.9%	5.9%	5.9%
Corporate income taxes	1.1%	1.1%	1.4%	1.3%
Other	1.4%	1.3%	1.2%	1.2%
Total Revenues	**16.4%**	**16.6%**	**17.0%**	**17.9%**

Source: Congressional Budget Office, Baseline Budget Projections as of March 6, 2020. Figures are annual averages for the projected ranges shown.

10.1.2 US Forecasting Practices

US governments use quantitative techniques mixed with judgement to arrive at final forecasts. Methods include:

- Time series methods (ARIMA, moving averages, exponential smoothing) — most common at local government level.
- Causal and causal-like methods (simulations, econometric or systems of statistical models) used by nations and large subnational jurisdictions.
- Decomposition and mixed approaches — total forecast is the deterministic sum of separately-forecast tax components.
- Consensus forecasting — used in part to remove politics from the process.
- Regression-style forecast models — adjusting future values of independent variables to reflect a policy change and comparing against a current-policy baseline.

10.2 United Kingdom | Macroeconomic Model

The Office for Budget Responsibility (OBR) was created to examine and report on the UK's public finances. It publishes five-year forecasts for the key components of the public finances twice a year, alongside each Budget and Autumn Statement. Its macroeconomic model is a simplified representation of the economic activity recorded in the National Accounts, with equations broken into three groups:

- Accounting identities — equations specifying the identities and definitions in the National Accounts (e.g., real GDP equals the sum of consumption, investment, government spending, and net trade).
- Behavioural (econometric) equations — econometrically estimated equations based on economic theory and statistical analyses of past behaviour (e.g., household spending responding to real incomes, interest rates, and wealth).
- Technical relationships — calibrated relationships based on economic theory or broad historical trends, and stylised forecasting assumptions.

10.2.1 Receipts

National Insurance contributions and other receipts are estimated using the receipts group, which contains only technical relationships and 'imposed' variables estimated using individual tax models run by HM Revenue & Customs. The group feeds into household disposable income and the basic price adjustment within the macroeconomic model.

10.3 Italy | Macroeconomic Models

Italy's Parliamentary Budget Office (UPB) endorsed the 2019-2020 macroeconomic forecasts presented in the Update of the Economic and Financial Document (NADEF). Italy uses four main models:

- ITEM (Italian Treasury Econometric Model): 371 variables (247 endogenous), 36 behavioural equations, 211 identities. Used for both forecasting and macroeconomic impact assessment.
- IGEM (Italian General Equilibrium Model): A medium-scale Dynamic General Equilibrium (DGE) model with New Keynesian characteristics. Incorporates a heterogeneous labour market.
- QUEST III with R&D: A Dynamic Stochastic General Equilibrium (DSGE) model developed by the European Commission for analysing structural reforms and shocks.
- MACGEM-IT: A static Computable General Equilibrium (CGE) model for Italy, built to quantify disaggregated, direct, and indirect impacts of fiscal policies.

10.4 Australia | Base Plus Growth

Australia's government prepares macroeconomic and fiscal forecasts using a range of modelling techniques including macro-econometric models, spreadsheet analysis, and accounting frameworks, supplemented by survey data, business liaison, professional opinion, and judgement.

- Real GDP forecasts factor in exchange rates, interest rates, commodity prices, and domestic-international linkages.
- Nominal GDP forecasts are subject to additional uncertainty from domestic prices and wages, import prices, and global commodity prices.
- Tax receipts are estimated using a 'base plus growth' methodology: the last known outcome is used as the base, to which estimated growth rates are applied to generate current and future-year estimates.

10.5 Germany | Macroeconomic Models

Germany's forecasting and budget process allows for a variety of independent or semi-independent institutions to participate in fiscal policy and budget planning. The key institutions include: the Council of Economic Experts (est. 1963); the independent economic institutions producing the Joint Economic Forecast (est. 1950); the Advisory Board to the Federal Ministry of Finance (est. 1950); the Working Party on Tax Revenue Forecasting (est. 1955); and the Independent Advisory Council (est. 2013) to the Stability Council.

The German annual budget process operates in three 'arcs':

- Arc 1 (November to April): Council of Economic Experts report; internal Ministry of Finance tax revenue forecast; Stability Programme preparation.
- Arc 2 (April to October): Joint Economic Forecast commissioned from independent institutes; official Macroeconomic Forecast published; Working Party tax revenue forecast.
- Arc 3 (October to December): Second Joint Economic Forecast; final official Macroeconomic Forecast; budget finalisation.

Table 3: Germany — Methodological Aspects of Revenue Estimation

Estimation Technique	Relevant Step	Model / Technique Features	Assumptions
Macroeconomic forecast	Before each tax revenue estimate	Iterative-analytic approach: several partial models used in the system of national accounts. Potential GDP estimated using Output Gaps Working Group (EU EPC) models.	Technical assumptions for oil/commodity prices, exchange rates and interest rates
Tax revenue estimate	Basis for drafting and finalising budget	Estimate based on macroeconomic forecast and time series analysis	Macroeconomic forecast; estimates on fiscal impact of discretionary tax measures
Fiscal impact of discretionary tax measures	Basis for tax revenue estimate and for drafting/finalising budget	Microsimulation models based on tax statistics; calculations based on macroeconomic assumptions	Macroeconomic forecast

Source: German Draft Budgetary Plan 2020; German Ministry of Finance.

10.6 Netherlands, Japan, New Zealand | Econometric Models

The Netherlands, New Zealand, and Japan reportedly make use of econometric models where the relationship between revenues and economic indicators is directly estimated using regression analysis. Microsimulation methods are used for individual taxpayers (households or corporations) — this approach is used in the Netherlands and the United States for forecasting income taxes, and in the United Kingdom and the United States for corporation tax. For less important taxes or those with a weak relationship to macroeconomic variables (e.g., certain excise duties), trend-extrapolation or formal time-series analysis is employed.

11 National Revenue Forecasting: NIPFP Studies

This section reviews three National Institute of Public Finance and Policy (NIPFP) working papers that provide important methodological and empirical context for revenue forecasting in India and in Telangana specifically.

11.1 Kerala's Revenue Potential Projections for 1990–95

Kerala's performance in resource mobilisation during the Sixth Plan period fell far below expectations, with the shortfall in the Balance from Current Revenue (BCR) reaching 68% of original estimates — the highest among Southern States. A NIPFP study revealed the failure was attributable to:

- Shortfall in revenue receipts rather than excessive spending.
- Sluggish growth in own revenue rather than of Central transfers.
- The shortfall was greater in own tax revenues, though non-tax revenue (particularly interest receipts) registered a larger percentage gap between targets and actuals.
- Deceleration in revenue growth was the main factor responsible for poor BCR contribution to Plan resources.

11.1.1 Tax Revenue Projections for the Eighth Plan: Methodology

The exercise first assessed the tax potential of the State using the Tax System Approach, rather than merely relying on past growth rates. The Tax System Approach involves:

1. Identifying the tax base for each individual tax by the appropriate legislative provision.
2. Estimating the size of the tax base for the relevant forecast year.
3. Multiplying the tax base estimate by the applicable tax rate structure to obtain the revenue potential estimate.
4. Applying an estimated compliance rate to the revenue potential to obtain the projected revenue collection.

11.1.2 Regression Method for Estimating Potential

As a complementary approach, regression analysis was used to estimate revenue potential. For operational purposes, tax and non-tax revenue are examined separately. The table below summarises the regression method used for Kerala's medium-term projections for 1990-91 to 1994-95:

Method	Projection Period	Uses
Regression method	Medium term (5 years)	Improving the collections from certain taxes individually (stamp duties and registration fees, tax on vehicles, profession tax, entertainment tax). Non-tax revenue sources appear to provide good scope for resource mobilisation on an equitable basis.

The exact method adopted consists of two sets. The First Set: for each (group of) tax, a function is postulated between the tax revenue and the immediate tax base or proxies, tested through regression equations on historical data, then refined until the best statistical fit is obtained. Forecast values of the tax base are then plugged into the preferred regression equation to yield projections for 1990-95.

The Second Set: a variant that scales the First Set upward by the maximum tax effort factor observed within the reference period. This gives the upper limit of potential tax revenue — what Kerala could raise if its tax base exploitation were as intensive as in its peak year since 1970-71.

11.1.3 Individual Taxes

The table below details the revenue-to-base functions postulated, tested, and adopted for each individual tax in Kerala. The choice of equations was guided by statistical properties. For each tax, four standard functional forms were tried: linear, log-linear, and two semi-log specifications.

Table: Kerala Individual Taxes — Assumptions and Regression Functions

Individual Tax	Key Assumptions	Regression Function
Land & Agricultural Taxes	1. Both land revenue and agricultural income tax depend on the productivity of land. 2. SDP from the primary sector adopted as the suitable proxy base. 3. Taken as indicator of productivity given relative constancy of area under cultivation.	`LAGTAX = f (SDPP)` `Where: LAGTAX = revenue from land and agricultural taxes; SDPP = SDP from the primary sector.`
Stamp Duties & Registration Fees	1. Proximate bases are the number and value of documents or instruments of transfer attracting these levies. 2. Property values expected to have some relationship with average household income level. 3. Urbanisation is a major factor underlying spiralling of property values in urban areas.	`SDRF = f (PCSDP, URB)` `Where: SDRF = revenue from stamp duties and registration fees; PCSDP = per capita SDP; URB = urbanisation (%).`
State Excise Duties	The base is quite obviously the consumption of different kinds of liquor. Available data on liquor consumption was not fully dependable.	`EXCD = f (PCSDP, POPN)` `Where: EXCD = revenue from excise duties; POPN = population.`
Sales Tax	1. This tax has a fairly wide base, comprising consumption as well as production. 2. NSS data is the primary source for consumption, but inadequate for a time series. 3. In Kerala, agricultural sector output is largely cash crops; sales tax from agriculture cannot be ignored. 4. Tax on goods sold by producers directly across state boundaries cannot be levied under GST Act.	`GST = f (SDPAFF, SDPMFG, ORB, BANKS)` `Where: GST = general sales tax revenue; SDPAFF = SDP from agriculture, fishing & forestry; SDPMFG = SDP from manufacturing; BANKS = no. of commercial bank branches.`
Motor Vehicles Tax	Tax revenue specified as a function of the number of different types of motor vehicles on the road.	`MVT = f (GOODS, BUS, TAXI, 2WH, AUTO, OTHER)` `Where: MVT = motor vehicle tax revenue; GOODS = goods vehicles; BUS = stage carriages; TAXI = taxis; 2WH = two-wheelers; AUTO = auto rickshaws.`
Electricity Duty	Tax revenue taken to be a function of consumption of electricity by different types of consumers.	`ED = f (DOM, COMM, INDL, IRR, PUBSER, BULK)` `Where: ED = electricity duty revenue; DOM = domestic; COMM = commercial; INDL = industrial; IRR = irrigation; PUBSER =`

Individual Tax	Key Assumptions	Regression Function
		`public services; BULK = bulk supply.`
Local Bodies		
Entertainment Tax (Local Bodies)	Base for this tax is the number of cinema halls and/or their seating capacity. Information in time-series form was unavailable; proxy bases used.	`ENTTAX = f (PCSDP, POPN)` `Where: ENTTAX = total entertainment tax revenue.`
Profession Tax (Local Bodies)	A direct tax; base taken to depend on SDP arising mainly in the non-primary sector.	`PROTAX = f (NPSDP)` `Where: PROTAX = profession tax revenue; NPSDP = SDP from non-primary sector.`

Note: SDP = State Domestic Product; PCSDP = Per Capita SDP; URB = Urbanisation (%); POPN = Population; NPSDP = Non-Primary SDP. Data on 'other taxes' showed minimal and erratic collections; no regression was attempted. Property tax was excluded due to paucity of base data.

11.2 Tax Revenue Forecasts of Central, State and Local Governments in India (2000–01 to 2004–05)

This NIPFP working paper (Diwan Chand et al., 2000), prepared for the Eleventh Finance Commission, examined revenue projections for the Union, state governments, and local bodies over a five-year plan period.

11.2.1 Objective

The primary objective was to project central and state tax revenues for the medium-term fiscal framework (2000–01 to 2004–05) to assist the Eleventh Finance Commission in its award determinations. The methodology combines multiple approaches at different stages of projection:

- Central tax revenues projected separately for corporate income tax, non-corporate income tax, excise duty, customs, and other taxes.
- State tax revenues projected for agricultural taxes, sales tax, state excise, stamp duties and registration fees, motor vehicle taxes, electricity duty, and other taxes — for all 25 individual states.
- At the local level, estimates of aggregate revenue from local taxes, and for property tax and octroi separately, are made for 11 states for which data are available.

Table: Projection Period and Method by Government Level

Government Level	Projection Period	Method
Central Government	Medium term (5 years)	Log-linear regression
State Governments	Medium term (5 years)	Estimation based on one or more determining variables (GSDP-based buoyancy equations)
Local Bodies	Medium term (5 years)	Assumed growth rate method (trend growth rates adjusted by ratio of actual to assumed all-India GDP growth)

11.2.2 Projection of Central Tax Revenue

Revenue-forecasting and reassessment exercises by the Finance Commissions generally took into account historical patterns of revenue mobilisation and trends in tax policy changes. The underlying assumption was that past trends in the tax bases, and the effect of structural changes in the tax system, would remain more or less constant. The table below summarises the forecast approaches adopted by successive Finance Commissions, followed by the methodology adopted in the present study:

Table: Forecast Approach Adopted by Finance Commissions — 7th to 10th FC

Finance Commission	Key Assumptions & Approach
Till 7th Finance Commission	1. Long-term and recent revenue growth trends; elasticities of revenues of different taxes with respect to Net National Product (NNP) as well as NNP originating in the non-agricultural sector and manufacturing sector. 2. Changes in the tax structure in recent years, including concessions given. 3. Considered somewhat higher rates of revenue growth than assumed by the Central Government.
8th Finance Commission	1. Partial elasticities with respect to income and price variables; also took note of specific developments such as the mopping up of ₹1,000 crore through bearer bonds. 2. Structural change in industrial production that affected growth of revenue from Union excise duties. 3. Assumed growth rates: income tax 6.5%, corporation tax 7.5%, Union excise duties 7%, customs duties 7%.
9th Finance Commission (First Report)	1. Assumed 5% growth in GDP and prices; projected growth rates: income tax 8.5%, corporation tax 13.09%, Union excise duties 13.4%, customs duties 20.34%. 2. These growth rates were applied to the base of 1987-88 (RE). Care was taken to ensure overall tax revenue growth rate of 15% for 1989-90 and a long-term rate of 14.5%.
9th Finance Commission (Second Report)	1. Assumed 6% GDP growth during 1990-95 and 5% rise in prices.
10th Finance Commission	1. In projecting revenues for 1995-2000, the base year (1994-95) figures were first re-estimated to account for sensitivity of forecast values to the base year. 2. Next, buoyancy coefficients for individual taxes (income tax, corporate tax, customs, Union excise duties) were estimated with respect to GDP at current market prices for the sample period 1980-81 to 1990-91.
Present Study Methodology: The determinants method is adopted, linking tax revenue projections to GDP growth rates and expected inflation rates for the 9th Five-Year Plan period (6.5% target / 7.4% optimistic). Log-linear regression: Log(X) = α + β × log(Y) + u, where X = tax revenue, Y = GDP/GSDP proxy base, β = tax buoyancy coefficient.	

The present study adopts the determinants method using log-linear regression to link tax revenue projections to GDP growth and inflation. Three sets of forecasts are produced: (I) Baseline — based on past trends of GDP series; (II) Target — based on the 9th Five-Year Plan target GDP growth rate of 6.5% per annum; and (III) Optimistic — based on the optimistic growth rate of 7.4%.

11.2.3 Decomposition of Tax Revenue Growth into Base and Price Effects

A key methodological contribution of the paper is the decomposition of total tax revenue growth into a base effect (growth attributable to real expansion of the tax base) and a price effect (growth attributable to inflation). If taxes are charged on the money value of the bases (as is mostly the case in India), the estimated tax-buoyancy equation enables straightforward segregation of these effects:

$$x^{g} = \beta \times (y^{Rg} + p)$$

Where: x^g = tax revenue growth; y^{Rg} = real GDP growth; p = rate of inflation; β = buoyancy coefficient. Thus, the growth of the tax yield is the sum of the growth of the proxy base in constant prices and the rate of inflation, together weighted by the buoyancy coefficient.

11.2.4 Projection of State Government Tax Revenues

Two basic methods are used for projecting state government tax revenues: (a) Trend projection method — utilises past values of the same variable being projected, sometimes incorporating information about the future (e.g., policy decisions). (b) Estimation based on one or more determining variables — requires estimating or postulating a behavioural relationship between variables, then estimating future values of the dependent variable based on independent estimates of the determining variables. The table below summarises the method, assumptions, accuracy considerations, and applications:

Table: State Government Tax Revenue Projection — Method, Assumptions, Accuracy & Uses

Method	Assumptions	Accuracy of Forecasts	Application / Uses
Estimation based on one or more determining variables (GSDP-based double-log elasticity)	1. Tax revenues for each state determined by the State Domestic Product (SDP/GSDP). Relationship assumed to be double-log so that the elasticity coefficient of tax revenues with respect to GSDP can be directly obtained. 2. Given projected values of GSDP, tax revenues are projected using these estimated elasticity coefficients.	1. Data series of actual economic variables often exhibit irregularities that do not allow simple universal application of a chosen method. 2. In several cases, estimated equations were found inadequate in terms of explanatory power, or buoyancy coefficients turned out to be statistically insignificant.	1. Data series carefully examined to identify discrete changes signifying structural changes. 2. Identified irregularities taken into account by excluding outliers or using a dummy variable for such observations. 3. Where significant compound growth rates could not be obtained, simple average annual growth of relevant variables was used.

Key Issues in State-Level Projections

1. For Jammu & Kashmir, buoyancy coefficients could not be estimated due to non-availability of GSDP figures.
2. In other cases, irregularities in the entire data series of tax revenues precluded estimation of statistically significant buoyancy coefficients.
3. In all such cases, the alternative trend projection method was adopted.
4. In most such cases, statistically significant compound growth rates could not be obtained; simple average annual growth of relevant variables was used instead.
5. The base year figure for projections is a critical parameter. While 1997-98 is used as the base in general, care was taken to ensure base figures are not outliers — where they appeared too high or low, a 3–5 year average was used as the base year value.

11.2.5 Projection of Local Government Tax Revenues in Selected States

Local government tax projections were prepared for selected states using property tax and octroi/local body tax data. The paper notes data limitations at the local level and recommends institutional data collection improvements.

11.3 Tax Revenue Efficiency of Indian States: Stamp Duty and Registration Fees

This NIPFP working paper (A. Sri Hari Nayudu, 2019) measures tax capacity and tax effort of stamp duty and registration fees for 16 major Indian states from 2001 to 2014 using Stochastic Frontier Analysis (SFA).

Background: Following GST implementation, states lost jurisdiction over many taxes. The major own-revenue-yielding taxes remaining in the post-GST regime are excise duty and stamp duty and registration fees. Stamp duty constitutes the third or fourth most important source of own tax revenue for state governments, with the all-India average ranging between 7% and 14% of own tax revenue (Mukherjee, 2013).

Key findings:

- Bihar is operating at high efficiency levels; Odisha and Jharkhand are operating with low efficiency.
- The gap between predicted tax revenue and frontier tax revenue is most pronounced in Gujarat, Rajasthan, Tamil Nadu, Punjab, and West Bengal.
- State governments need to focus on stamp duty policy changes and relevant determinants to improve efficiency.

11.3.1 Stochastic Frontier Analysis (SFA)

The SFA model specifies the following log-linear relationship:

$$\ln y_{it} = \alpha + \beta x_{it} + v_{it} - u_{it}$$

Where: y_{it} and x_{it} represent the dependent variable (tax revenue) and independent variables for state i at time t; β's are vectors of parameters; v_{it} is the random disturbance (two-sided); u_{it} is the technical inefficiency term (non-negative, representing the gap between actual and frontier tax revenue). Statistical independence between v_{it} and u_{it} is assumed, with $u_{it} > 0$.

Different distributional assumptions on u_{it} lead to different SFA model classes, including: cross-sectional models, panel-data models, time-variant models, and True Fixed Effects (TFE) and True Random Effects (TRE) models.

11.3.2 Tax Revenue Prediction to 2022

After estimating efficiency levels, total stamp duty and registration tax revenues were predicted to 2022 under the following conditions:

- All independent variables grow at a linear rate.
- Good monsoon is assumed throughout the prediction period.
- FRBM is followed by all states throughout the period.

The gap between predicted and frontier tax revenue is largest for Gujarat, Rajasthan, Tamil Nadu, Punjab, and West Bengal. A key limitation of the study is the inability to incorporate exact stamp duty rates and corruption-related variables due to data unavailability.

12 Telangana Revenue Forecasting: Sample Workings

A government budget is a document prepared by the government presenting its anticipated revenues and proposed spending for the coming financial year. Tax Revenue forms a substantial portion of revenue receipts for most Indian states, comprising State's Own Tax Revenue (SOTR) and each state's share in Union Taxes (devolution). Revenue forecasting helps governments plan, control, and track expenditure commitments and maintain fiscal discipline.

12.1 State's Own Tax Revenue (SOTR)

SOTR includes revenues earned through Sales Tax, State Excise Duty, SGST, Land Revenue, and Stamps & Registrations. States that generate more revenue on their own are less dependent on devolution and central grants.

Telangana has managed to increase the share of SOTR to 83.61% of total tax revenue for the first time in FY 2020-21. Although the state previously had a higher share in earlier years, the share was in the 70s. This increase reflects both a rise in SOTR (by approximately ₹10,000 crore) and a fall in devolution of union tax revenue. Telangana is among the few states showing a year-on-year increase in the SOTR share.

12.2 Objective

- Forecast Telangana Tax Revenue and SOTR for FY 2020-21 and FY 2021-22 using data from FY 2014-15 to FY 2019-20.
- Validate the methods by forecasting FY 2018-19 and FY 2019-20 using data from FY 2014-15 to FY 2017-18 and comparing with actuals/revised estimates.

12.3 Forecasting Methods Applied

Table 4: Forecasting Methods Applied to Telangana Revenue Data

S.No.	Method	Model / Formula
1	**Tax Buoyancy / Tax Elasticity**	Tax Revenue (Next Year) = (1 + Tax Buoyancy × % Change in GSDP) × Tax Revenue (This Year)
2	**Macroeconomic / GDP-Based Model**	Logarithmic Regression
3	**Time Series**	ARIMA — Autoregressive Integrated Moving Averages
4	**Consensus Forecasting**	Multiple Linear Regression

12.4 Projection of Telangana Tax Revenue

12.4.1 Total Tax Revenue Projections

Table 5: Telangana Tax Revenue — Actual vs Forecast by Method (₹ Crore)

Year	Actual Tax Revenue (₹ Cr)	Tax Buoyancy	Log. Regression	ARIMA	Consensus
2018-19 (Actual)	83,235	84,073	95,087	74,628	83,207
2019-20 (RE)	87,315	95,043	1,15,692	68,232	87,316
2020-21 (BE)	1,02,027	91,609	1,12,303	97,283	1,02,035
2021-22 (Projected)	—	1,19,285	1,44,055	1,07,250	1,12,080

For FY 2018-19 and FY 2019-20, Tax Buoyancy and Logarithmic Regression overestimated tax revenue; ARIMA underestimated; Consensus Forecasting slightly underestimated by ₹28 crore in FY 2018-19 and overestimated by only ₹0.43 crore in FY 2019-20.

For FY 2020-21, discrepancies in results arise because the data is based on actuals rather than projections, which introduces some model instability.

12.4.2 State's Own Tax Revenue (SOTR) Projections

Table 6: Telangana SOTR — Actual vs Forecast by Method (₹ Crore)

Year	Actual SOTR (₹ Cr)	Tax Buoyancy	Log. Regression	ARIMA	Consensus
2018-19 (Actual)	64,674	69,905	75,092	59,176	64,702
2019-20 (RE)	71,328	71,903	91,882	53,859	71,327
2020-21 (BE)	85,300	78,703	90,711	79,736	82,357
2021-22 (Projected)	—	1,01,995	1,09,157	88,144	90,759

For FY 2018-19 and FY 2019-20, Tax Buoyancy and Logarithmic Regression overestimated SOTR; ARIMA underestimated; Consensus Forecasting overestimated by ₹28 crore in FY 2018-19 and underestimated by ₹1 crore in FY 2019-20 — errors well within acceptable bounds.

12.5 Forecast Errors

The tables below present the forecast errors (Actual minus Forecast) for Tax Revenue and SOTR. Positive values indicate the model underestimated actual collections; negative values indicate overestimation. Green values indicate errors below ₹1,000 crore; amber values indicate larger errors.

Tax Revenue Forecast Errors (₹ Crore)

Table 7a: Tax Revenue Forecast Errors

Year	Actual (₹ Cr)	Tax Buoyancy Error	Log. Regression Error	ARIMA Error	Consensus Error
2018-19 (Actual)	83,235	−838	−11,852	+8,607	+28
2019-20 (RE)	87,315	−7,728	−28,377	+19,084	−0.43
2020-21 (BE)	1,02,027	+10,417	−10,276	+4,744	−9

State's Own Tax Revenue Forecast Errors (₹ Crore)

Table 7b: SOTR Forecast Errors

Year	Actual (₹ Cr)	Tax Buoyancy Error	Log. Regression Error	ARIMA Error	Consensus Error
2018-19 (Actual)	64,674	−5,231	−10,418	+5,498	−28
2019-20 (RE)	71,328	−575	−20,554	+17,469	+1
2020-21 (BE)	85,300	+6,597	−5,411	+5,564	+2,943

In comparison with the other models, Consensus Forecasting most accurately estimates both Tax Revenue and SOTR, achieving minimum errors across all validation years. This model can reliably be used to predict data points 2–3 years ahead. A limitation is that it requires prior forecasting of all independent variable values before generating the revenue projection.

13 Organisation of Revenue Forecasting

13.1 Responsibilities and Challenges of Revenue Target Setting

Successful revenue forecasting and target setting requires effective assignment of roles and coordination between three key organisational units:

- The Tax Policy Unit (TPU) in the Ministry of Finance (MoF) — prime responsibility for tax policy, revenue forecasting, and target setting in the context of macroeconomic and budget planning, including debt and reserve fund planning. Also responsible for tax expenditure policy and monitoring of tax administration performance.
- The Research and Planning Unit of the Revenue Administration (RA) — prime responsibility for administering the tax laws and meeting revenue collection targets. Primary source for micro-level tax assessment, collection, compliance, and administration data.
- Statistical and economic management agencies — key contributors providing economic data, models, and macroeconomic plans and forecasts. Where the central bank acts as the government's banker, it is also a key source of information on final tax deposits.

The MoF must manage coordination and cooperation between agencies, which is critical for reliable and credible forecasts. A crucial element is the sharing of tax data, models, and key assumptions between MoF and RA, so that both parties can interpret the causes of deviations from target collections and take shared ownership of outcomes.

13.2 Sources of Failure in Revenue Forecasting

In practice, there are many sources of failure in revenue forecasting, including:

- Failure of the RA to collect data critical to tax analysis and forecasting, due to poor tax form design or failure to establish a complete tax database.
- Legal and/or bureaucratic control problems in data access and sharing. For example, tax laws may not give MoF officials access to administrative data for taxes under their responsibility; alternatively, the RA fails to collect or refuses to share critical tax details.
- Failure by the MoF to conduct evidence-based analysis or share models and assumptions with the RA and other key agencies. If targets are inconsistent with macro forecasts, the RA may resort to holding back refunds or pressuring larger taxpayers — strategies that deteriorate taxpayer compliance over the long run.

Computerisation is key to establishing tax databases and enabling data sharing, analysis, and forecasting. However, internal computer networks within the RA and MoF have often suffered from: (i) arithmetic and other errors in tax returns; (ii) limited data fields keyed into computer systems; (iii) sampling of returns for data capture; and (iv) data entry errors. E-returns and e-filing are progressively reducing these limitations.

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Annexure A: Sources of Information by Country

The following country-specific references document or discuss the revenue forecasting approaches reviewed in Chapter 10 of this report.

A.1 Austria

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A.10 New Zealand

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A.11 United Kingdom

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A.12 United States

Auerbach (1999); Congressional Budget Office (1995): Budget Estimates: Current Practices and Alternative Approaches; CBO (1998): Projecting Federal Tax Revenues; CBO (2006): CBO's Policies for Preparing and Distributing Its Estimates and Analyses; CBO (2007): The Uncertainty of Budget Projections; Joint Committee on Taxation (1992): Discussion of Revenue Estimation Methodology and Process.

Annexure B: NIPFP-Related Working Paper Series

The following NIPFP working papers are relevant to the analysis presented in this report and are recommended for further reading:

- Ghosh, A., & Chakraborty, L. (2019). Analysing Telangana State Finances: Elongation of Term to Maturity of Debt to Sustain Economic Growth. NIPFP Working Paper No. 288.
- Rao, M. G. (2017). Public Finance in India in the Context of India’s Development. NIPFP Working Paper No. 219.
- Mukherjee, S. (2019). Whether States Have Capacity to Sustain Projected Growth in GST Collection During the Compensation Period? NIPFP Working Paper No. 275.

Annexure C: Contact Information

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